

Research Article

Advanced Design and Comparative Analysis of Methanol Production Routes from CO₂ and Renewable H₂: via Syngas vs. Direct Hydrogenation Processes

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In this study, we propose two advanced processes, i.e., advanced syngas-to-methanol (AS2M) and direct CO₂-to-methanol (DC2M) processes, for methanol production using waste CO₂ and renewable H₂ as feedstock. The AS2M process produces methanol from syngas after converting CO₂ to CO via reverse water-gas shift and separating CO from a mixture through pressure swing adsorption. In comparison, the DC2M process directly synthesizes methanol from CO₂ through direct hydrogenation. Different unit processes and recycles were synthesized to determine the optimal process configurations and operating conditions for maximizing the methanol production rate. The technoeconomic and environmental capabilities of the two proposed processes were analyzed with four evaluation criteria (carbon and energy efficiencies, CO₂ reduction, and unit production cost) and compared with conventional CO₂-to-methanol processes. A sensitivity analysis for various utilities and H₂ supply strategies was performed to evaluate the economic feasibility and environmental benefits of the proposed systems with different design options. Our findings showed that both processes could achieve improved technical, economic, and environmental performances compared to conventional processes. In particular, the DC2M process showed the lowest unit production cost at \$1.02/kg because of its simple configuration and mild operating conditions.

1. Introduction

CO₂ emission from the excessive use of fossil fuels is considered an acute and severe environmental problem. The Intergovernmental Panel on Climate Change (IPCC) emphasized the importance of limiting the global temperature rise to less than 1.5°C to solve this issue [1]. Therefore, many researchers have also focused on solving this problem by proposing different CO₂ utilization technologies, such as enhanced oil recovery [2], CO₂-enhanced coalbed methanol recovery [3], and carbon capture and utilization (CCU) [4–7]. CCU is one of the promising strategies as it produces high-value materials and chemicals, such as carbonates, alcohols, and hydrocarbons, while mitigating CO₂ emissions [8, 9]. A number of studies on CCU have been extensively reported in the literature across various research areas such as catalyst/material synthesis and development [10, 11], reactor design and kinetic

study [12, 13], process integration and intensification [14, 15], and technoeconomic and life-cycle assessments [16, 17]. In detail, Aresta et al. assessed the CO₂ conversion of potential biocatalysts and corresponding energy products [18]. Rafiee et al. described various pathways, development progress, and process benchmarks on CO₂ capture and utilization [15]. González-Garay et al. studied a molecule-to-planet level assessment for the green methanol production processes using CO₂ and renewable H₂ [19]. Kim et al. integrated solar-based fuel processes with CO₂, evaluated the environmental impacts through life-cycle assessment, and accordingly proposed a new framework for CO₂ utilization [20–22]. Do et al. proposed 72 CO₂-to-fuel pathways and evaluated their technical, economic, and environmental performances by utilizing process systems engineering-centric approaches, such as process synthesis, simulation, and optimization [23]. As a practical project launched in 2017, the ALIGN-CCUS project has

focused on optimizing the carbon capture, usage, and storage (CCUS) chain, specifically life cycle assessments, public acceptance, and exploring implementation options [24]. Its findings inform future research, guide policy decisions, and support efforts toward achieving low-carbon industrial growth. With their concept, Moser et al. [25] conducted a comparative technoeconomic analysis on complete chains of dimethyl ether (DME) production, while Troy et al. [26] focused on the life cycle assessment of both DME and polyoxymethylene dimethyl ethers (OMEs). Albrecht et al. proposed a standardized methodology to assess the economic viability of sustainable fuels derived from the Fischer-Tropsch (FT) synthesis [27]. Moreover, Von der Assen et al. developed a framework for conducting reliable environmental evaluations of CCU processes using life cycle assessment methods [28].

Of the various CO₂-based products, methanol is widely used in energy industries because it serves as a raw material for other chemicals, as a fuel itself, or as a fuel additive (blended with gasoline) [29]. Thus, researchers have focused on methanol-oriented studies, from catalyst development to process design and system analysis. Typically, methanol is produced using fossil fuels. It is synthesized using syngas (a mixture of CO and H₂), which is produced by converting raw materials (such as natural gas or coal) via stream reforming or gasification processes [30, 31]. However, the captured CO₂ has been used as a cofeed with natural gas in dry reforming [29] or with H₂ in the reverse water-gas shift for syngas intermediates [32–34] due to the increasing demand for methanol in the energy sectors and the global attention on CO₂ mitigation. Joo et al. proposed a new methanol process using CO₂ and H₂ as feeds, which is known as the CAMERE process (“CARbon dioxide hydrogenation to form methanol via a Reverse water-gas shift reaction”) [35]. In this process, CO₂ and H₂ are first converted into CO and H₂O through the reverse water-gas shift reaction (RWGS: $\text{CO}_2 + \text{H}_2 \leftrightarrow \text{CO} + \text{H}_2\text{O}$), and CO obtained from RWGS and additional H₂ are resynthesized to produce methanol via a methanol synthesis process (MS: $\text{CO} + 2\text{H}_2 \leftrightarrow \text{CH}_3\text{OH}$, $\text{CO}_2 + 3\text{H}_2 \leftrightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$). In the paper, the unreacted gases in methanol synthesis are purged out of the reactor to maintain operating conditions and feed pressure. However, this proposed process scheme causes losses of valuable H₂ and the reemission of CO₂ into the atmosphere. Therefore, Kim et al. added a recycled stream from a methanol reactor to the RWGS reactor and thus modified the CAMERE process; this resulted in a higher methanol yield [21]. Although these two papers have evaluated the economics involved in their proposed methanol processes using CO₂ and H₂, there is a need to increase the compatibility of the CAMERE process compared to the state-of-the-art methanol production technologies, such as direct CO₂ hydrogenation.

In contrast to the CAMERE process that produces syngas as an intermediate, direct CO₂ hydrogenation, an up-to-date technology, produces methanol from CO₂ without producing syngas [36]. Thus, direct hydrogenation has a simple process scheme due to the absence of syngas production and reduces the utility consumption required for reactor or separator operations. An et al. examined the Cu/Zn/Al/Zr catalyst activities

and developed corresponding kinetic models suitable for direct hydrogenation [37]. Kiss et al. [38, 39] and Do and Kim [40] proposed new process schemes appropriate for direct CO₂ hydrogenation and conducted a technoeconomic analysis. In addition, Cho et al. developed a new knowledge-based assessment platform for early-stage screening of CO₂ direct hydrogenation catalysts based on the simulation results [41].

This study aims to propose new methanol synthesis routes from CO₂, advanced syngas-to-methanol (AS2M), and direct CO₂-to-methanol processes (DC2M), which exhibit improved performance compared to the existing routes. This study also comparatively analyzed technoeconomic and environmental performances of the two proposed routes using various evaluation criteria. A detailed explanation of the AS2M and DC2M processes can be found in Section 2, and the process modeling and simulation are discussed in Section 3. We evaluated and compared their technical, economic, and environmental performances with those of the original CAMERE process proposed by Joo et al. [35] and the modified CAMERE process proposed by Kim et al. [21]. For this, carbon efficiency (CE), energy efficiency (EE), carbon reduction (CR), and unit production cost of methanol (UPC) were used as key performance indicators (Section 4). Moreover, a sensitivity analysis of utility sources and H₂ production technologies is performed in Section 5, and the paper is concluded in Section 6.

2. Problem Statements

2.1. Description of CO₂ to Methanol Production Process.

Figure 1 shows four different process configurations for methanol production from CO₂ and H₂. It is assumed that CO₂ is captured from flue gas using amine-based absorption, and H₂ is produced via water electrolysis using renewable electricity. Meanwhile, utilities (electricity and heat) required to operate unit processes are assumed to be generated from conventional sources. Figure 1(a) indicates the process scheme of the original CAMERE process (henceforth referred to as the o-CAMERE process) initially proposed by Joo et al. [35]. As mentioned, CO₂ and H₂ are fed to the RWGS reactor, which converts them to both CO and H₂O. The resulting stream consists of produced CO, unreacted H₂, and CO₂ that are moved to a methanol synthesis (MS) reactor to produce methanol (mainly via CO hydrogenation). The raw methanol synthesized by the MS reactor is then fed to the methanol purification (MP) unit to obtain a high-purity methanol product (more than 99% purity). Figure 1(b) represents the modified CAMERE process developed by Kim et al. (henceforth referred to as the m-CAMERE process) [21]. The m-CAMERE process has an analogous configuration compared to the o-CAMERE process, including RWGS, MS, and MP. The distinct difference between the two CAMERE processes is that in the m-CAMERE process, the unreacted gases at the MS reactor are separated and recycled back to the RWGS reactor to increase the yield of methanol. Although the two CAMEREs are rigorously simulated in this paper using a commercial process simulator, both processes are used only as references for evaluating the impacts of advanced processes, as shown in

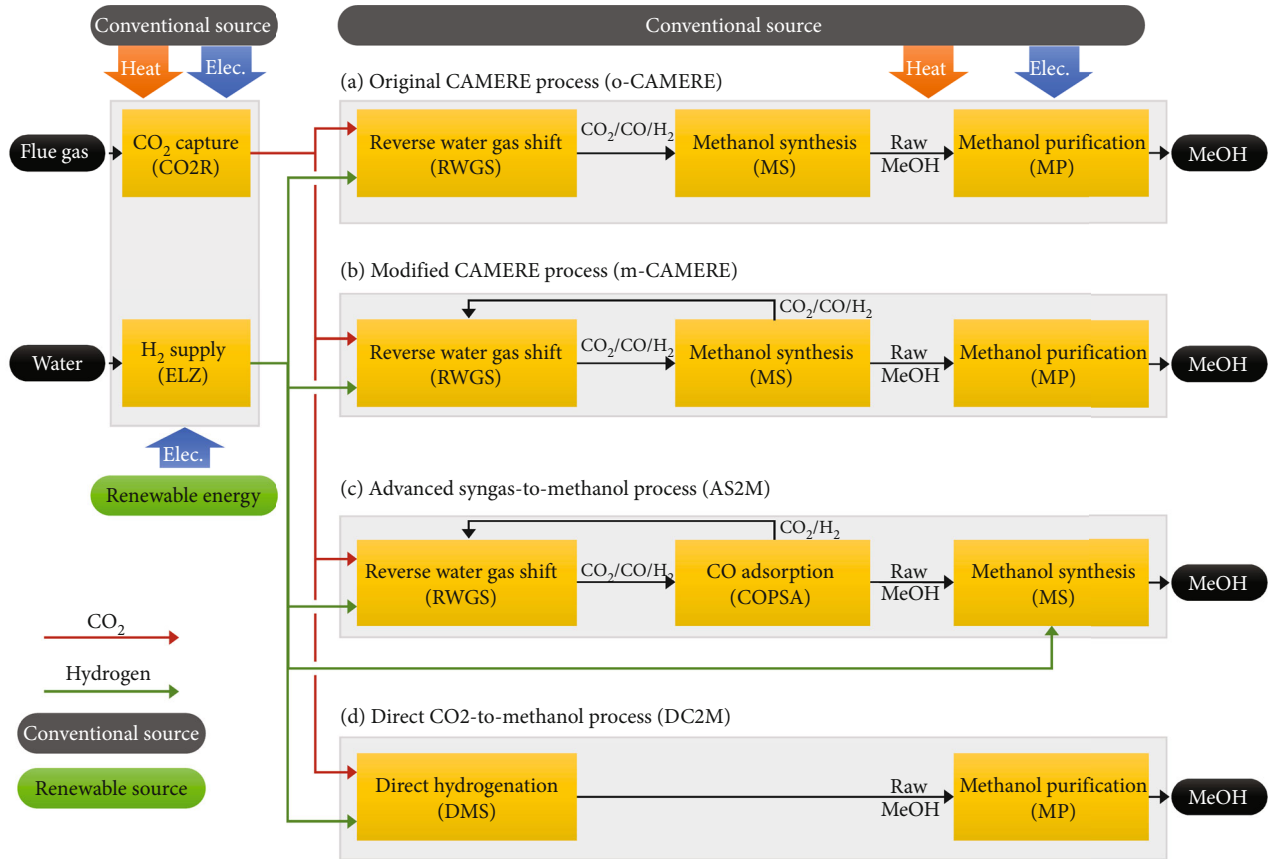


FIGURE 1: Process scheme of four CO₂ to methanol production processes: (a) Original CAMERE (o-CAMERE), (b) modified CAMERE (m-CAMERE), (c) advanced syngas-to-methanol (AS2M) process, and (d) direct CO₂-to-methanol (DC2M) process.

Figures 1(c) and 1(d). Figures 1(c) and 1(d) show the two advanced methanol processes proposed in this paper: AS2M and DC2M. The AS2M process is a modified CAMERE process, which includes CO pressure swing adsorption (CO PSA), i.e., selectively captures CO from the RWGS outlet gas (mixture CO₂/CO/H₂). The captured CO and outsourced H₂ are fed into the MS process, while CO₂ and H₂ are recycled into the RWGS process to increase CO production. The H₂ flow was manipulated to achieve the optimal H₂/CO ratio for methanol production. Thereby, high-purity methanol is fully obtained in the MS reactor with CO/H₂ feed, stating that additional MP process is not required. Meanwhile, the DC2M process directly converts CO₂ to methanol under a Cu-based catalyst in one-spot conversion [37, 39]. Figure 1(d) shows that CO₂ and H₂ are fed to a CO₂ direct MS (DMS) reactor. Compared to the MS reactor, the DMS performs a lower one-pass conversion to methanol due to the high stability and low reactivity of CO₂. The unreacted CO₂ and H₂ are recycled to the DMS reactor to obtain a higher methanol yield, resulting in more extensive recycle streams and larger equipment sizes. The raw methanol is then purified in the MP process to achieve over 99% purity of methanol.

2.2. Evaluation Criteria. Four metrics such as carbon efficiency, energy efficiency, CO₂ reduction, and unit production cost were used to evaluate the technical, environmental, and economic feasibilities of the processes.

These four metrics were also used to compare the advantages of the AS2M and DC2M processes against the o-CAMERE and m-CAMERE processes.

Carbon efficiency (CE) represents the ratio of carbon in the product to carbon in feeds. It evaluates the number of carbons converted to the final product, as expressed in the following:

$$CE (\%) = \frac{C_{MeOH}}{C_{CO_2}}, \quad (1)$$

where C_{MeOH} and C_{CO_2} denote the total number of carbons in methanol and CO₂, respectively, calculated as shown in the following:

$$C_{MeOH} = \tau_{MeOH}^C F_{MeOH}, \quad (2)$$

$$C_{CO_2} = \tau_{CO_2}^C F_{CO_2}, \quad (3)$$

where F_{MeOH} and F_{CO_2} denote the flow rates of produced methanol and reacted CO₂ (kmol/h), and τ_{MeOH}^C and $\tau_{CO_2}^C$ denote the amount of carbon element in methanol and CO₂ (kmol of carbon/kmol of compounds). Energy efficiency (EE) is calculated using the chemical energy of methanol and the total utility consumed in the process as forms of electricity or heat. As shown in Equation (4), the EE represents how efficiently the energy is used to convert feed (i.e., CO₂) to product (i.e., methanol).

$$EE (\%) = \frac{E_{\text{MeOH}}}{E_{\text{elect}} + E_{\text{heat}}}, \quad (4)$$

where E_{MeOH} denotes the chemical energy of methanol (MJ of methanol/yr), which is calculated as shown in Equation (5). E_{elect} and E_{heat} denote consumed utilities in the process as forms of electricity and heat (MJ of electricity or heat/yr), respectively.

$$E_{\text{MeOH}} = \delta \pi_{\text{MeOH}} M_{\text{MeOH}} F_{\text{MeOH}}, \quad (5)$$

where δ denotes the operability of the process, which is assumed to be 8,000 hours/year, π_{MeOH} denotes the low heating value of methanol (i.e., 22.7 MJ/kg MeOH), and M_{MeOH} denotes the molar weight of methanol (i.e., 32.04 kg/kmol). The CO₂ reduction (CR) effect may occur in the CO₂ utilization processes as CO₂ is used as a feedstock. Nevertheless, CO₂ can also be emitted during the process via direct emission and indirect emission. Direct emission refers to CO₂ emissions as a form of vent-out gas during the operation, while indirect emission indicates CO₂ emissions that are caused by utility production from conventional sources. Accordingly, the net CR of the methanol production process is defined in this paper to evaluate the cradle-to-gate effect of CO₂ in methanol production, as shown in Equation (6). In detail, a positive CR states that the amount of CO₂ used as a feed is greater than the amount of CO₂ released during the process. In other words, CO₂ concentration is reduced by utilizing the given process.

$$CR (\text{kg CO}_2/\text{kg MeOH}) = \frac{\delta \{M_{\text{CO}_2} F_{\text{CO}_2} - (\text{DE} + \text{IE})\}}{M_{\text{MeOH}} F_{\text{MeOH}}}, \quad (6)$$

where M_{CO_2} denotes the molar weight of CO₂ (i.e., 44.01 kg/kmol). DE denotes the direct emission of CO₂ during the process (kg CO₂/h), and IE represents the indirect emission of CO₂, and it is calculated as follows:

$$\text{IE} = e_{\text{elect}} E_{\text{elect}} + e_{\text{heat}} E_{\text{heat}}, \quad (7)$$

where e_{heat} and e_{elect} denote the CO₂ emission factors of heat and electricity (kg of CO₂/MJ of heat or electricity), respectively, which are calculated based on energy mix. In this study, the unit production cost (UPC) of methanol is a key performance indicator to evaluate the economic capability of the methanol production processes. It is calculated using the total production cost (TPC) and the annual production rate of methanol, as expressed in the following equation:

$$\text{UPC} (\$/\text{kg MeOH}) = \frac{\text{TPC}}{\delta M_{\text{MeOH}} F_{\text{MeOH}}}. \quad (8)$$

Therein, TPC (\$/yr) is composed of capital cost (CC), operating cost (OC), and byproduct credit (BC), in consideration of capital charge factor (a) (Equation (9)). Here, CC denotes the equipment purchase costs obtained from the Aspen Process Economic Analyzer. OC denotes the expenses incurred during the process operation and is composed of fixed-operating cost (FC), which comprises labor cost, plant overhead, maintenance

cost, and variable operating cost (VC), which comprises raw material cost and utility cost, as shown in Equation (10). The FC is usually equivalent to a fixed percentage of the CC, while the VC is calculated, as shown in Equation (11).

$$\text{TPC} = a\text{CC} + \text{OC} - \text{BC}, \quad (9)$$

$$\text{OC} = \text{FC} + \text{VC}, \quad (10)$$

$$\text{VC} = \sum_{k \in K_u} \theta_k E_k + \sum_{k \in K \setminus K_u} \theta_k A_k, \quad (11)$$

where θ_k denotes a unit price of utility k , E_k denotes the amount of utilities $k \in K_u$ (i.e., electricity and heat) consumed (MJ/yr), and A_k denotes the amount of other utilities, except for electricity and heat $k \in K \setminus K_u$, such as cooling water, adsorbent, absorbent, and ionized water.

BC is a credit from selling O₂ (byproduct) produced through water electrolysis. As stated in Equation (12), it is calculated by multiplying the unit price of O₂ (θ_{O_2}) and the amount of O₂ produced along with H₂ (F_{O_2}). The capital charge factor (a) is used to amortize the capital cost into the annual cost, which is calculated based on economic plant lifetime (n) and interest rate (s), as shown in Equation (13).

$$\text{BC} = \theta_{\text{O}_2} F_{\text{O}_2}, \quad (12)$$

$$a = \frac{s(1+s)^n}{(1+s)^n - 1}. \quad (13)$$

3. Modeling and Simulation

The rigorous process models of the four processes were developed using Aspen Plus V11 [42]. The processes were simulated using stoichiometry and kinetics obtained from the literature [21, 40]. A Soave-Redlich-Kwong was chosen as a property method. To maximize methanol production, the operating conditions (temperature and pressure) for each unit process and the optimal H₂ flows were optimized by using the optimization tool built into Aspen Plus. By default, Aspen Plus assigns successive quadratic programming (SQP) algorithm to solve small or large-scale optimization problems [43]. The mass and energy balance and sizing and costing data were obtained for further evaluation. Economic performance was evaluated by using the simulation results obtained by the Aspen Process Economic Analyzer. It is assumed that heat exchange networks are not considered in process simulation, which will be further improved in future work.

3.1. Unit Process Modeling. The four main methanol production processes are composed of multiple unit processes, including conversion and separation processes. There are a total of seven unit processes such as CO₂ recovery (CO2R), CO pressure swing adsorption (COPSA), MP, Electrolyzer (ELZ), RWGS, MS, and DMS. Specifications, including simplified PFD, chemical equations, kinetics, and operating conditions, for each unit process are summarized in Table 1.

TABLE 1: Unit process specification and operating conditions.

Process	Simplified PFD ¹	Operating conditions T (°C) P (bar)	Description
CO ₂ recovery (CO2R)		25–120 1–15	(i) Amine absorption for CO ₂ capture from flue gas (ii) Absolvent–monoethanolamine (MEA) (iii) Target spec–CO ₂ purity: >96%, CO ₂ recovery: >85% [21]
CO pressure swing adsorption (COPSA)		60 9	(i) Adsorbent– θ -Al ₂ O ₃ supported with Cu/Cu ⁺ (ii) Density–0.9 g/cm ³ , replacement time–2 years (iii) Target spec–CO purity: ~99%, CO recovery: ~90% [44] (iv) Hierarchical model is used in Aspen Plus
Methanol purification (MP)		Top: 1 Bottom: 102–107	(i) Target spec–MeOH purity: ~99%, MeOH recovery: ~98% [40]
H ₂ electrolyzer (ELZ)		25 1	(i) Type–alkaline electrolyzer [45] (ii) Efficiency–50 kWh of electricity/kg of H ₂
Reverse water gas shift (RWGS)		400–1,200 1–20	(i) Chemical reaction $\text{CO}_2 + \text{H}_2 \leftrightarrow \text{CO} + \text{H}_2\text{O}, \Delta H_{298\text{K}} = 41 \text{ kJ/mol}$ (ii) Kinetics ² $r_1 = \frac{k_2 P_{\text{CO}_2} (1 - P_{\text{H}_2\text{O}} P_{\text{CO}} / k_3 P_{\text{H}_2} P_{\text{CO}_2})}{(1 + K_1 P_{\text{H}_2}^{0.5} + K_2 P_{\text{H}_2\text{O}} + K_3 P_{\text{H}_2\text{O}} / P_{\text{H}_2})}$ (iii) Catalyst–Cu/ZnO/Al ₂ O ₃ [23]

TABLE 1: Continued.

Process	Simplified PFD ¹	Operating conditions T (°C) P (bar)	Description
Methanol synthesis (MS)		150–300 50–75	(i) Chemical reaction $\text{CO} + 2\text{H}_2 \leftrightarrow \text{CH}_3\text{OH}, \Delta H_{298\text{K}} = -91 \text{ kJ/mol}$ $\text{CO}_2 + \text{H}_2 \leftrightarrow \text{CO} + \text{H}_2\text{O}, \Delta H_{298\text{K}} = 41 \text{ kJ/mol}$ $\text{CO}_2 + 3\text{H}_2 \leftrightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}, \Delta H_{298\text{K}} = -49 \text{ kJ/mol}$ (ii) Kinetics² $r_1 = k_2 \frac{K_{\text{CO}} (f_{\text{CO}} f_{\text{H}_2}^{1.5} - f_{\text{CH}_3\text{OH}} / K_{\text{H}_2}^{0.5})}{(1 + K_{\text{CO}} f_{\text{CO}} + K_{\text{CO}_2} f_{\text{CO}_2}) (f_{\text{H}_2}^{0.5} + K_{\text{H}_2\text{O}} f_{\text{H}_2\text{O}} / K_{\text{H}_2}^{0.5})}$ $r_2 = k_3 \frac{K_{\text{CO}_2} (f_{\text{CO}_2} f_{\text{H}_2}^{1.5} - f_{\text{H}_2\text{O}} f_{\text{CH}_3\text{OH}} / K_{\text{H}_2}^{0.5})}{(1 + K_{\text{CO}} f_{\text{CO}} + K_{\text{CO}_2} f_{\text{CO}_2}) (f_{\text{H}_2}^{0.5} + K_{\text{H}_2\text{O}} f_{\text{H}_2\text{O}} / K_{\text{H}_2}^{0.5})}$ $r_3 = k_4 \frac{K_{\text{CO}_2} (f_{\text{CO}_2} f_{\text{H}_2} - f_{\text{H}_2\text{O}} f_{\text{CO}} / K_{\text{H}_2}^{0.5})}{(1 + K_{\text{CO}} f_{\text{CO}} + K_{\text{CO}_2} f_{\text{CO}_2}) (f_{\text{H}_2}^{0.5} + K_{\text{H}_2\text{O}} f_{\text{H}_2\text{O}} / K_{\text{H}_2}^{0.5})}$ (iii) Catalyst: Cu/ZnO/Al₂O₃ [23]
Direct hydrogenation for MeOH synthesis (DMS)		250–350 35–55	(i) Chemical reaction $\text{CO} + 2\text{H}_2 \leftrightarrow \text{CH}_3\text{OH}, \Delta H_{298\text{K}} = -91 \text{ kJ/mol}$ $\text{CO}_2 + \text{H}_2 \leftrightarrow \text{CO} + \text{H}_2\text{O}, \Delta H_{298\text{K}} = 41 \text{ kJ/mol}$ $\text{CO}_2 + 3\text{H}_2 \leftrightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}, \Delta H_{298\text{K}} = -49 \text{ kJ/mol}$ (ii) Kinetics² $r_1 = k_1 \left(\frac{K_{\text{CO}} f_{\text{CO}} f_{\text{H}_2}^{1.5} - (K_{\text{CO}} / K_A) f_{\text{CH}_3\text{OH}} f_{\text{H}_2}^{-0.5}}{K_{\text{H}_2}^{0.5} + K_{\text{H}_2\text{O}} f_{\text{H}_2\text{O}} / K_{\text{H}_2}^{0.5}} \right) / \text{ABS}$ $r_2 = k_2 \left(\frac{K_{\text{CO}_2} f_{\text{CO}_2} f_{\text{H}_2} - (K_{\text{CO}_2} / K_B) f_{\text{H}_2\text{O}} f_{\text{CO}}}{K_{\text{H}_2}^{0.5} + K_{\text{H}_2\text{O}} f_{\text{H}_2\text{O}} / K_{\text{H}_2}^{0.5}} \right) / \text{ABS}$ $r_3 = k_3 \left(\frac{K_{\text{CO}_2} f_{\text{CO}_2} f_{\text{H}_2}^{1.5} - \frac{K_{\text{CO}_2}}{K_C} f_{\text{H}_2\text{O}} f_{\text{CH}_3\text{OH}} f_{\text{H}_2}^{-1.5}}{K_{\text{H}_2}^{0.5} + K_{\text{H}_2\text{O}} f_{\text{H}_2\text{O}} / K_{\text{H}_2}^{0.5}} \right) / \text{ABS}$ $\text{ABS} = f_{\text{H}_2}^{0.5} + \frac{K_{\text{H}_2\text{O}}}{K_{\text{H}_2}^{0.5}} f_{\text{H}_2\text{O}} + K_{\text{CO}} f_{\text{CO}} f_{\text{H}_2}^{0.5} + K_{\text{CO}_2} f_{\text{CO}_2} f_{\text{H}_2}^{0.5} + \frac{K_{\text{CO}_2} K_{\text{H}_2\text{O}}}{K_{\text{H}_2}^{0.5}} f_{\text{CO}_2} f_{\text{H}_2\text{O}}$ (iii) Catalyst: Cu/Zn/Al/Zr [23, 40]

¹Simplified PFDs, except for the PFD of H₂ electrolyzer and MP, are adapted from Ref [23]. ²Information on the rate constant for each unit process can be found in relevant references.

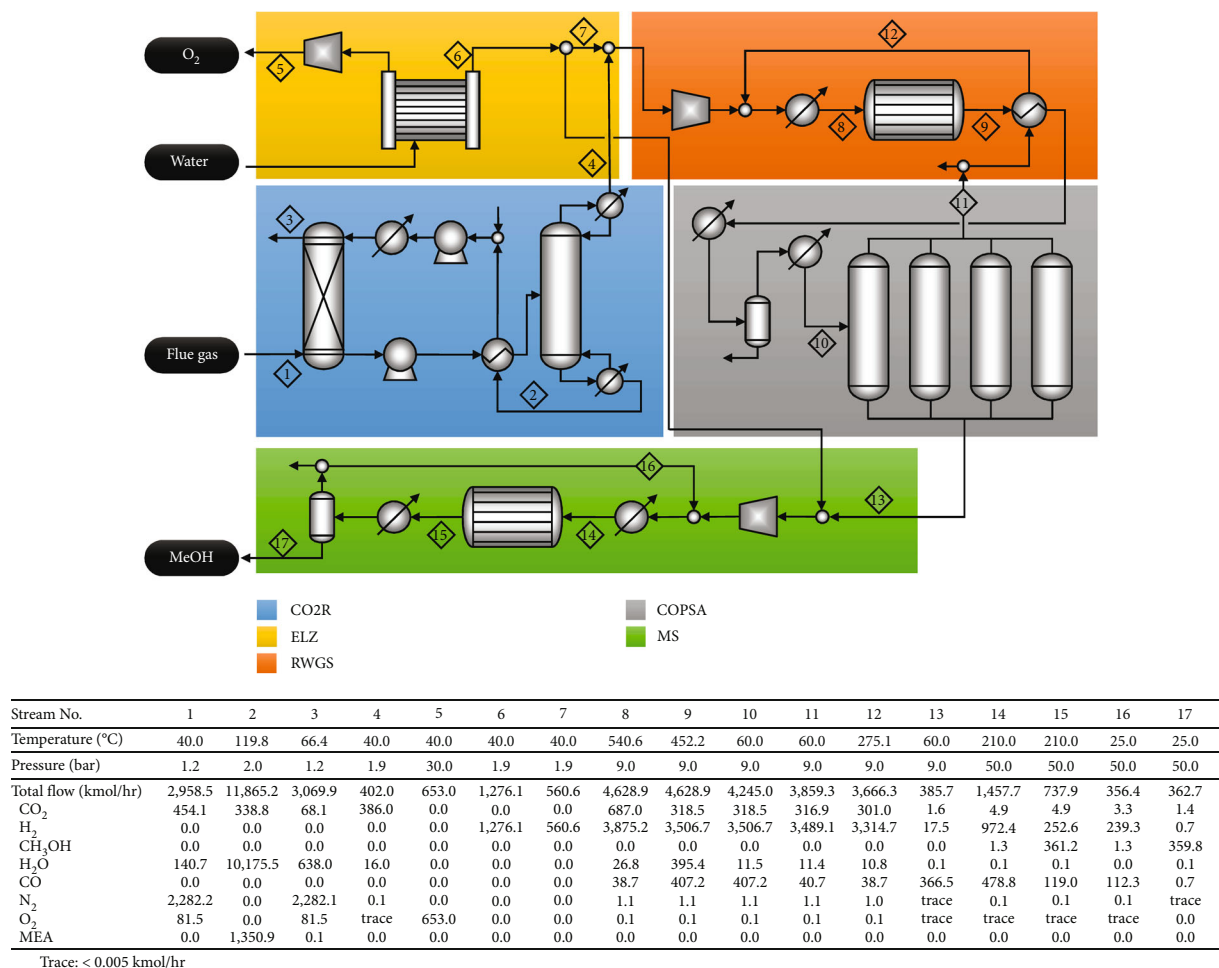


FIGURE 2: Process flow diagram and major stream information of the AS2M process.

3.2. Simulation Results. For all examined methanol production processes, CO₂ feedstock is assumed to be captured from the flue gas of the 30 MW power plant (flue gas flow rate: 2,959 kmol/h). The amine-based absorption using monoethanolamine (MEA) captures 386 kmol/h of CO₂ mixed with 16 kmol/h of H₂O. Pure CO₂ (96% purity) was supplied to the methanol production processes. Meanwhile, H₂ is produced via water electrolysis, with the amount required depending on the optimal operating condition of the unit processes.

Figure 2 presents the process flow diagram and major stream information (e.g., temperature, pressure, and mass flow) of the AS2M process. All captured CO₂ (386 kmol/h, stream #4) from the CO2R, H₂ produced by the ELZ (561 kmol/h, stream #7), and recycle stream from the COPSA (3,666 kmol/h, stream #12) are fed to the RWGS reactor. The RWGS reactor converts 369 kmol/h of CO₂ to CO, resulting in 53.6% CO₂ conversion (e.g., 369 kmol/h out of 687 kmol/h of CO₂). The outlet stream from the RWGS is fed to the COPSA, which captures 90% of CO from the mixture of CO₂, CO, H₂, and H₂O (e.g., 367 kmol/h of CO out of 407 kmol/h). The filtered CO was supplied to the MS for methanol production (stream #13), while other gases were recycled back to the RWGS reactor (stream

#11). In the MS reactor filled with Cu/ZnO/Al₂O₃ catalysts, the captured CO, additional H₂, and the inner recycle stream (mainly H₂, stream #16) were converted into methanol under optimal operating conditions and H₂/CO ratio (optimal ratio of H₂/CO in methanol synthesis ≈ 2.0). The MS reactor produced 361 kmol/h of methanol from 479 kmol/h of CO, resulting in a 75.4% CO conversion. H₂O and unreacted gas (a mixture of H₂/CO) were removed from the flash tank to improve the purification of methanol; 360 kmol/h of methanol with over 99% purity was obtained (stream #17).

Figure 3 presents the flow diagram of the DC2M process with key stream information. Similar to the AS2M, all captured CO₂ (386 kmol/h, stream #4), produced H₂ (1,147 kmol/h, stream #6), and an inner recycle stream (total 8,316 kmol/h, stream #10) are fed to the DMS reactor. The DMS reactor with Cu/Zn/Al/Zr catalysts directly converts CO₂ and CO to yield 360 kmol/h of net methanol. As mentioned before, the DMS process has the advantages of simple configuration and easy operation, but its drawback includes low methanol yield. It only has 17% CO₂ one-pass conversion, which requires a large amount of recycling to obtain a higher overall process conversion. As a result, the DC2M process has five times larger recycle flows than the feed flows

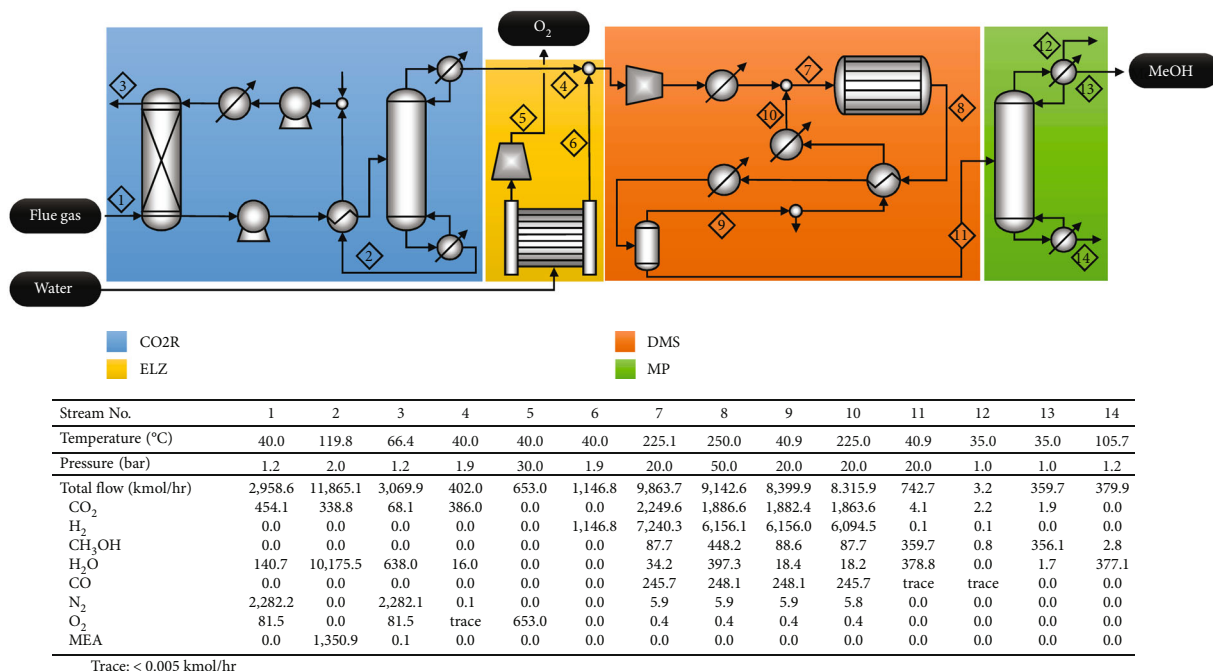


FIGURE 3: Process flow diagram and major stream information of the DC2M process.

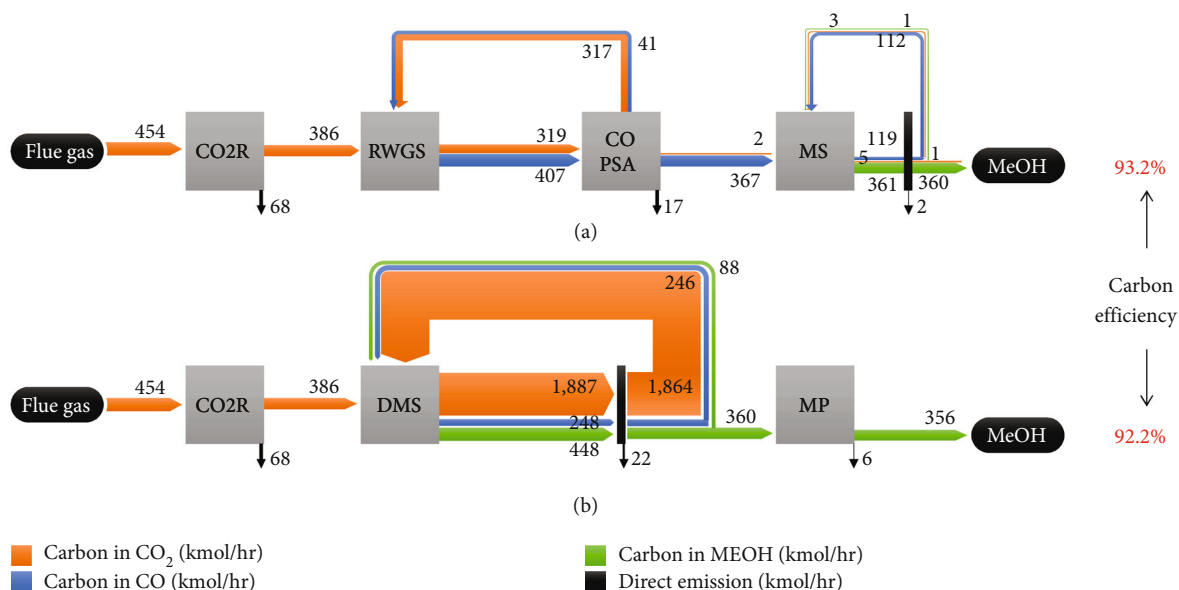


FIGURE 4: Carbon flow and carbon efficiency: (a) AS2M process and (b) DC2M process.

(i.e., CO₂+H₂ feeds: 1,549 kmol/h (streams #4 and #6) vs. recycles: 8,316 kmol/h (stream #10)). The stream in the flash tank is separated into gaseous compounds, such as CO₂, CO, and H₂, and liquid compounds, such as methanol and H₂O. The purified liquid flow is further treated in the MP (i.e., distillation column) to obtain approximately 99% pure methanol (stream #13). Notably, O₂ and H₂ are produced by the ELZ as byproducts (stream #5 in both cases), and O₂ is assumed to be sold to the external market for byproduct credits.

4. Technoeconomic and Environmental Impact Analysis

4.1. Carbon and Energy Efficiency Analysis. Figure 4 presents the carbon flows and carbon efficiency of both processes. The carbon flows can be broken down into the major components: CO₂, CO, methanol (MeOH), and direct emissions. As shown in Figure 4(a), 360 kmol/h of methanol is produced in the AS2M process using 386 kmol/h of CO₂ captured from the flue gas, indicating a carbon efficiency of 93.2%. On the other

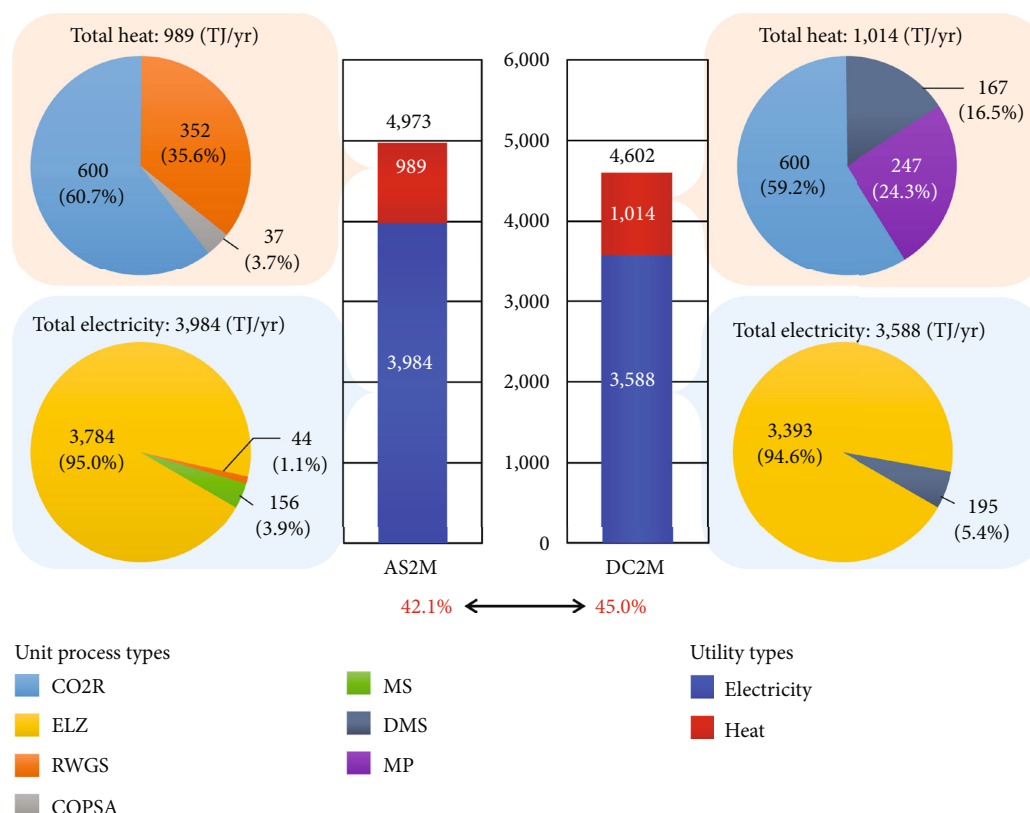


FIGURE 5: Energy consumption and energy efficiency of AS2M and DC2M processes.

hand, the DC2M process produces 356 kmol/h of methanol, indicating a carbon efficiency of 92.2% (see Figure 4(b)). Owing to the relatively better performances of RWGS, CO PSA, and MS, the AS2M process has a higher carbon efficiency than the DC2M process. As discussed in Section 3, the DC2M process has a large volume of recycle stream due to the low one-pass conversion of the CO₂ direct hydrogenation process, thus improving the yields of methanol.

Figure 5 presents the energy consumption and energy efficiency of both processes. The AS2M process consumes a total of 4,973 TJ/yr of utility, while the DC2M process requires 4,602 TJ/yr of utility. Compared with the AS2M process, the DC2M process involves a simple process scheme and easy operation, owing to low utility consumption. Electricity is the main utility consumed in both processes (i.e., 80% in the AS2M process and 78% in the DC2M process). Both processes consume approximately 95% of total electricity to decompose water into H₂ and O₂ in an ELZ. In the AS2M process, less than 5% of total electricity (i.e., 200 TJ/yr) is used to synthesize methanol in RWGS and MS. Likewise, in the DC2M process, 5.4% of electricity (i.e., 195 TJ/yr) is used for methanol production. CO₂R is the main contributor to heat consumption in both processes as the separation process of CO₂ and MEA solvent in a stripper requires a large amount of heat (see the simplified PFD of CO₂R in Table 1). RWGS also uses 352 TJ/yr of heat in the AS2M process because its reactor operates at a high temperature (~540°C). DMS reactor requires less heat (167 TJ/yr) than RWGS in the AS2M process due to its rela-

tively mild operating temperature (c.f., RWGS: ~540°C; DMS: ~225°C). Such low heat consumption of DMS in the DC2M process is offset by the large heat consumption of MP (i.e., 247 TJ/yr), which is required to improve the purity of methanol. Consequently, the DC2M process uses more heat utility than the AS2M process. A total of 360 kmol/h and 356 kmol/h of methanol produced in the AS2M and DC2M processes correspond to 2,094 TJ/yr and 2,072 TJ/yr, respectively; this results in energy efficiencies of 42.1% and 45.0% in the AS2M and DC2M processes, respectively. Notably, the AS2M process produces more methanol than the DC2M process (i.e., AS2M: 360 kmol/h, DC2M: 356 kmol/hr, see Figure 4), although it has lower energy efficiency due to its high energy consumption.

4.2. CO₂ Reduction Analysis. As shown in Figure 1, all unit processes, except for ELZ, use electricity and heat utilities obtained from conventional sources. Indirect CO₂ emissions from utility production and direct CO₂ emissions by vent-out or purge streams should be included in the evaluation of the environmental impacts of both AS2M and DC2M processes. Indirect CO₂ emissions are calculated based on Equation (7), and the CO₂ emission factors of the utilities are estimated to be 0.115 kg of CO₂/MJ of electricity and 0.062 kg of CO₂/MJ of heat, respectively. Here, it is assumed that heat is generated only from natural gas, while electricity is followed by the energy mix (e.g., coal (30.4%), natural gas (33.8%), nuclear (19.7%), oil (0.6%), renewable (14.9%), and others (0.6%)) [3, 46].

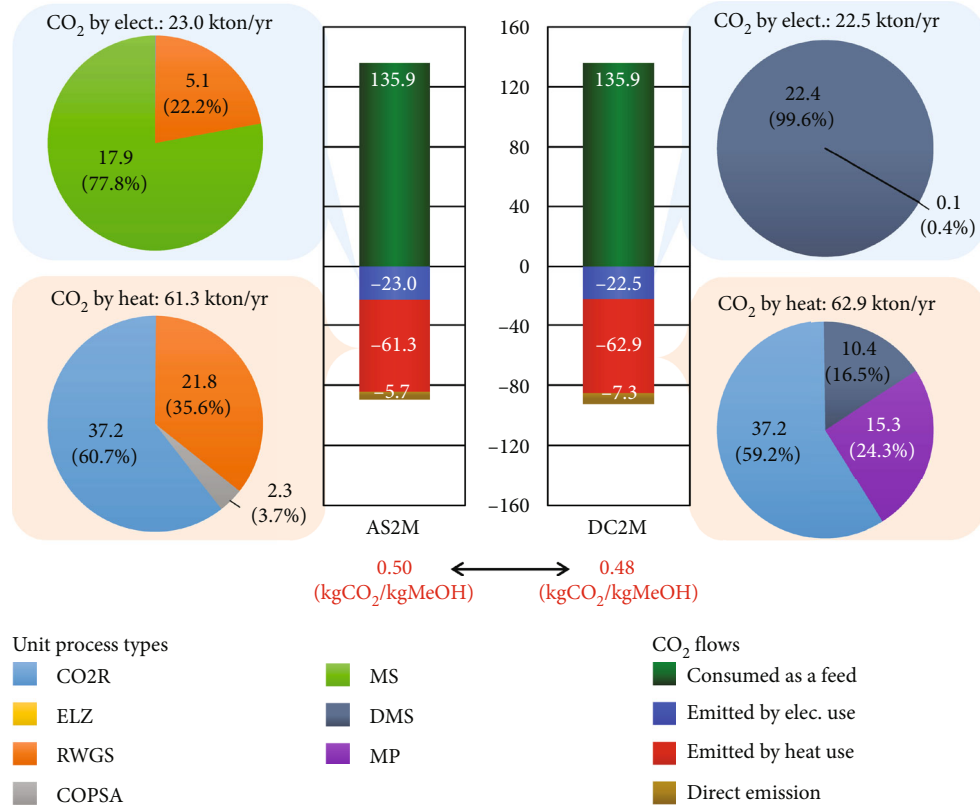
FIGURE 6: CO₂ emissions and reduction in the AS2M and DC2M processes.

Figure 6 shows the CO₂ emissions and CO₂ reductions of the AS2M and DC2M processes. The AS2M process emits a total of 90.0 kton of CO₂/yr, while the DC2M process emits a total of 92.7 kton of CO₂/yr. Heat consumption (AS2M: 68.1%; DC2M: 67.9% of the total emission), followed by electricity consumption (AS2M: 25.6%; DC2M: 24.3% of the total emission), and direct emissions (AS2M: 6.3%; DC2M: 7.8% of the total emission) are the main contributors to CO₂ emissions in both processes. Heat utility has a more negative impact on CO₂ emissions than electricity, even though both processes consume more electricity than heat (see Figure 5). This is because the electricity consumed in ELZ to decompose water is produced from renewable sources, which do not contribute to CO₂ emissions. Thus, the unit process that consumed more heat utilities emits more indirect CO₂ (such as CO2R, RWGS, and DMS). In addition, the DC2M process has larger direct CO₂ emissions than the AS2M process due to its large recycle flows (i.e., direct emission of AS2M: 5.7 kton/yr; DC2M: 7.3 kton/yr). As a result, both AS2M and DC2M processes are estimated to have CO₂ reductions of 0.50 and 0.47 kg CO₂/kg MeOH, respectively. These values indicate that, for instance, the AS2M process can reduce 0.50 kg of CO₂ by producing 1 kg of MeOH, mitigating the negative impact of CO₂.

4.3. Economic Analysis. The economic capability of the proposed two processes was evaluated using UPC, allowing a direct comparison of the economic viability of the different methanol production methods. In detail, the equipment cost

TABLE 2: General and economic parameters for economic evaluation.

Parameters	Values	Units	Refs.
General parameters			
Reference year	2018		
Interest rate	8	%	[23]
Economic plant lifetime	20	Years	[23]
Operability	8,000	Hour/yr	[3]
Economic parameters (utility and raw material prices)			
Electricity from grid	16.7	\$/kJ	[20]
Electricity from RES	22.2	\$/kJ	[47]
Heat from RES	13.8	\$/kJ	[23]
Low-pressure (LP) steam	1.9	\$/kJ	[20]
Medium pressure (MP) steam	2.4	\$/kJ	[20]
High-pressure (HP) steam	2.5	\$/kJ	[20]
Cooling water	0.03	\$/m ³	[20]
Deionized water	0.001	\$/ton	[45]
MEA for CO ₂ capture process	0.97	\$/kg	[48]
Adsorbent for CO capture process	50	\$/kg	[44]
Oxygen credit ¹	0.06	\$/kg	[49]

¹The profit obtained by selling byproducts (O₂) obtained through water electrolysis to the market.

of each unit such as the pump, heater, and distillation column is adopted from the Aspen Process Economic Analyzer

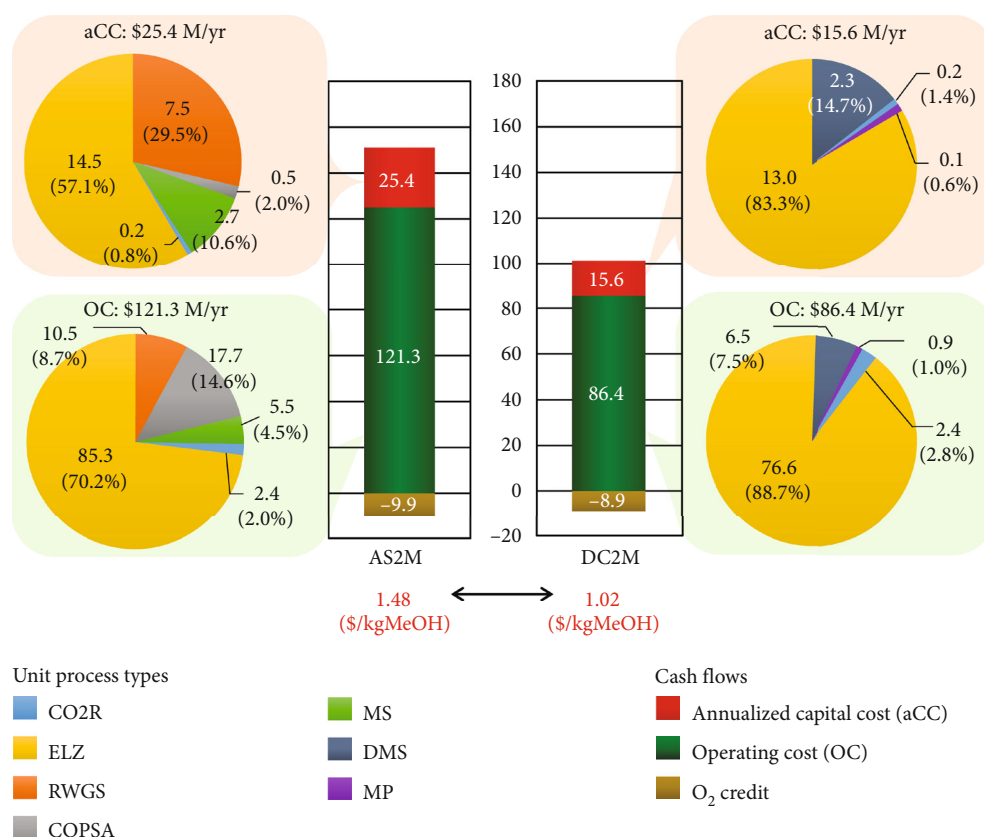


FIGURE 7: Cost contributions and unit production cost of AS2M and DC2M processes.

to estimate investment cost, and operating cost is calculated by using the amount of consumed utility and utility prices, as shown in Table 2.

Figure 7 presents the total production cost, which includes annualized capital cost (aCC), operating cost (OC), and O₂ credits, as well as the cost breakdown of the AS2M and DC2M processes. The TPC is \$136.8 M/yr for the AS2M process and \$93.1 M/yr for the DC2M process. The AS2M process has a more complex configuration than the DC2M process, which results in higher aCC and OC. In both processes, OC is approximately five times larger than aCC. In both processes, ELZ is the main contributor to aCC and OC. This is because the ELZ has a relatively higher CC than the other unit processes and requires a significant amount of electricity to decompose water. This implies that the economic feasibility of both processes can be greatly improved by lowering the CC and improving the conversion efficiency of the ELZ. COPSA has a significantly higher OC than the aCC (almost 35 times) because it uses an expensive adsorbent (see Table 2). As a result, the UPCs of the AS2M and DC2M processes are predicted to be \$1.48/kg MeOH and \$1.02/kg MeOH, respectively.

Figure 8 shows a standardization of five indicators such as carbon efficiency, energy efficiency, CO₂ reduction, profit, and water savings; their mean is zero and their standard deviation is one. Water savings are calculated based on the amount of cooling water consumed in the process (excluding the water consumed by the ELZ to produce H₂). The water savings are the reciprocals of the amount of cooling water. Meanwhile, the methanol profit is estimated by calcu-

lating the revenue from the methanol produced. The market price of methanol is assumed to be \$0.47/kg MeOH [45]. The detailed stream information of the o-CAMERE and m-CAMERE processes can be found in Figures S1 and S2 of the Supplementary Materials. In addition, the numerical results of the technical, economic, and environmental results of the two reference processes are also summarized in Table S1 in the Supplementary Materials. Compared to the conventional methanol processes, the AS2M and DC2M processes have a relatively higher carbon efficiency, energy efficiency, and methanol yield. The AS2M and DC2M processes reduce the net CO₂ level by effectively managing the involved processes. This result verified that the two proposed processes can provide environmental benefits. The economic profit of the AS2M process is less than that of m-CAMERE but more than that of o-CAMERE. As discussed in previous sections, the lower profit of the AS2M process is due to its complicated process scheme and tough operating conditions. On the other hand, the DC2M process is expected to have the largest profit among all four processes. Interestingly, the AS2M process does not have a MP unit, which requires a huge amount of water to purify methanol, and thus, it is expected to largely reduce water consumption compared to other processes.

5. Sensitivity Analysis and Comparison

In this section, scenario-based sensitivity analysis was performed to analyze the economic and environmental

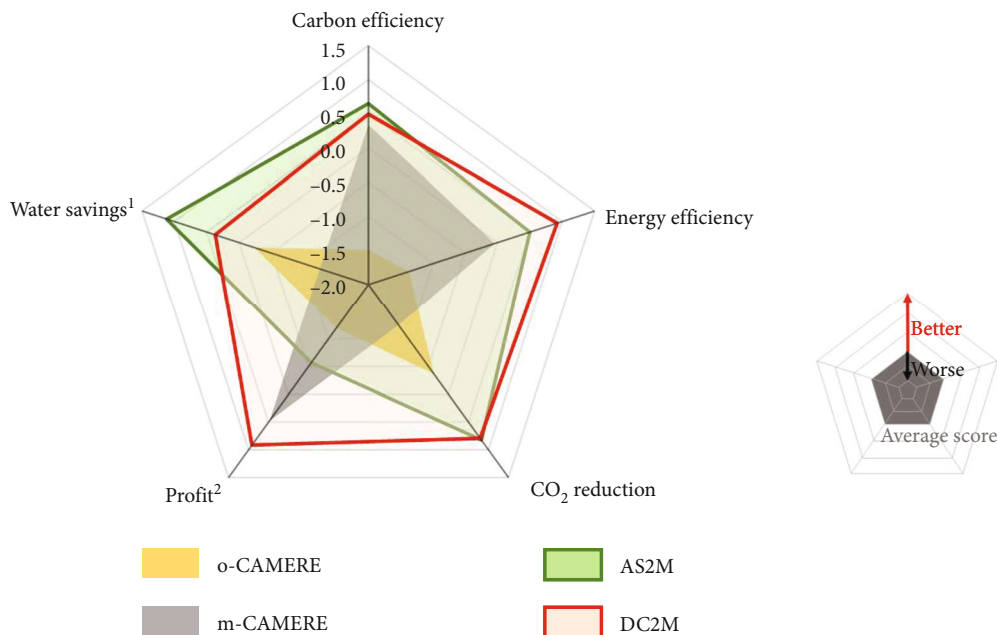


FIGURE 8: Holistic representation of sustainability of the four processes (larger values indicate better solutions in each dimension).
¹Water savings = 1/the amount of cooling water. ²Methanol profit = (revenue – cost)/yield of methanol = {(market price of methanol * yield of methanol) – cost}/yield of methanol}.

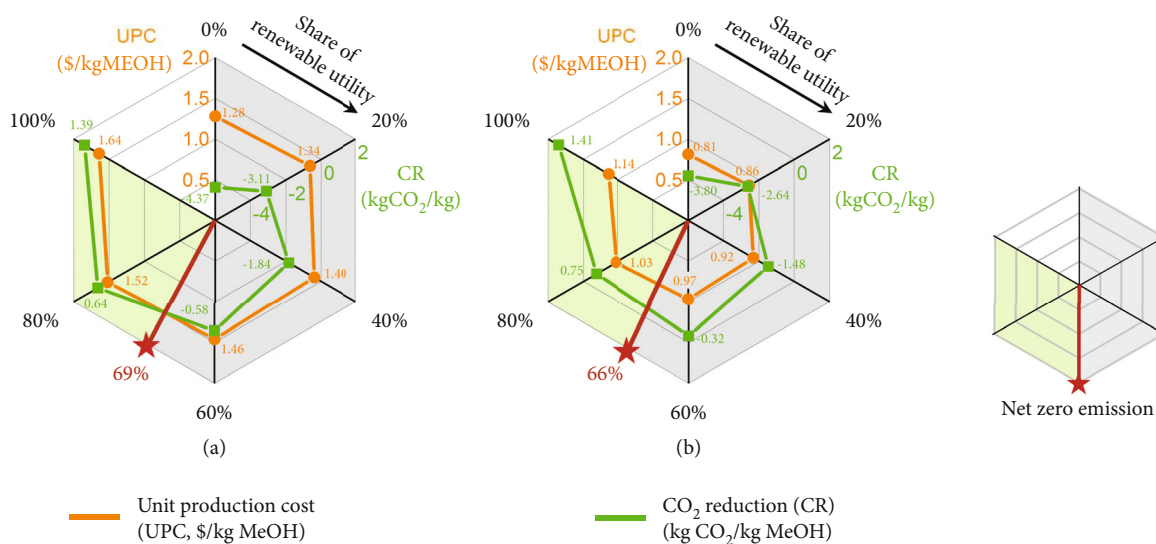


FIGURE 9: UPC and CR change with a share of renewable utility for the (a) AS2M process and the (b) DC2M process.

capabilities of the two processes (AS2M and DC2M) in various market conditions and different design options.

5.1. Utility Production Alternatives. As the baseline, the ELZ for H₂ production is assumed to be powered by electricity generated from renewable sources. The utilities consumed in the other subprocesses are assumed to be supplied from conventional sources. We analyzed the impact of various utility sources on the economic and environmental performances of both processes because the utility supply strategy is a critical factor in determining economic and environmental capabilities.

Figures 9(a) and 9(b) show the economic (orange spiral line) and environmental performances (green spiral line) of both AS2M and DC2M processes with a share of renewable utility, respectively. When conventional utilities (electricity and heat from fossil fuels) are fully used, even for water electrolysis, both processes can produce methanol at the lowest price (e.g., AS2M: \$1.28/kg MeOH; DC2M: \$0.81/kg MeOH). However, both processes indirectly emit a large amount of CO₂ during operation, resulting in a negative reduction effect (e.g., AS2M: -4.37 kg CO₂/kg MeOH; DC2M: -3.80 kg CO₂/kg MeOH). In contrast, the environmental capability of both processes can be improved by

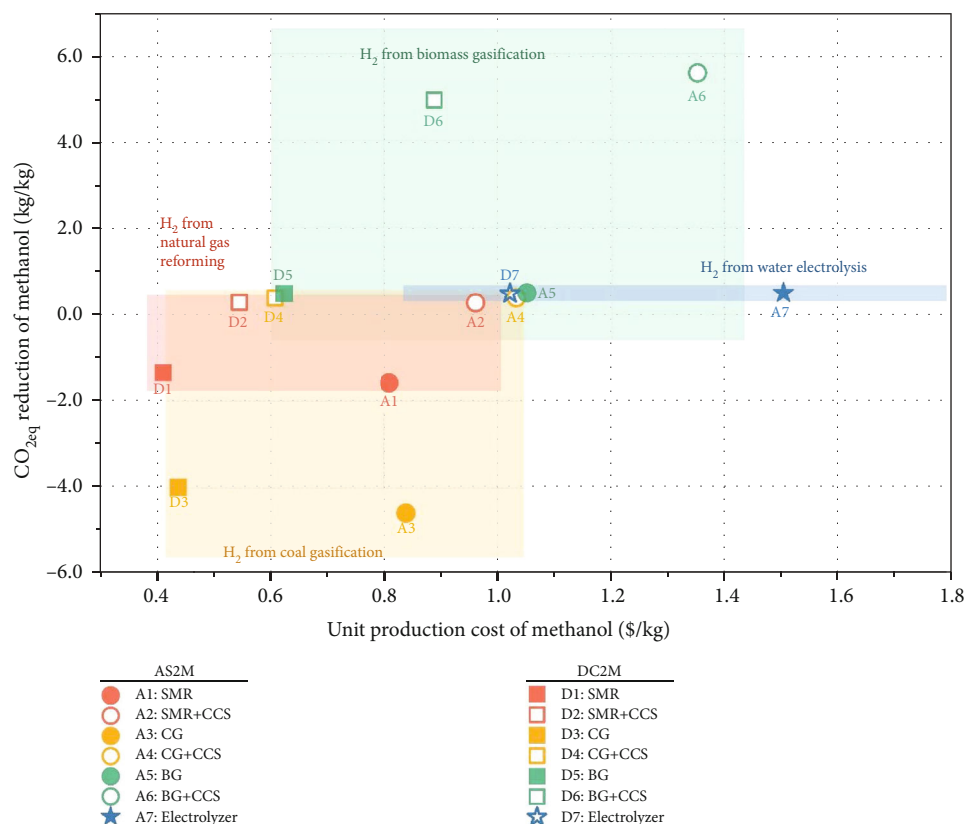


FIGURE 10: UPC and CO₂ reduction change with different hydrogen production technologies.

replacing all utilities with renewable utilities (CO₂ reduction; AS2M: 1.39 kg CO₂/kg MeOH; DC2M: 1.41 kg CO₂/kg MeOH). However, it cannot be an economically favorable alternative compared to using conventional utilities (UPC; AS2M: \$1.64/kg MeOH; DC2M: \$1.14/kg MeOH). Given the trade-off between the economic and environmental performances of the processes depending on the types of utilities, it is desirable to identify the ratio of renewable utilities used for net-zero emission processes. In this context, it is identified that both processes can produce methanol without emitting CO₂ (from a cradle-to-gate perspective) by partially utilizing renewable utilities. In detail, if at least 69% and 66% of the total utilities consumed in the AS2M and DC2M processes, respectively, can be replaced by renewable electricity and heat, both processes will produce carbon-neutral methanol, which makes the processes to be economically and environmentally favorable.

5.2. Hydrogen Production Alternatives. H₂ obtained from ELZ is the most critical factor in determining the economic feasibility of the proposed processes. H₂ can be produced from natural gas, coal, and biomass via different technologies (e.g., reforming or gasification) at a low price, but these processes have negative CO₂ emission. In this section, the economics and environmental impacts of the AS2M and DC2M processes, which are coupled with different H₂ production technologies (e.g., coal gasification, biomass gasification, and steam methane reforming), were evaluated. Figure 10 represents the UPC and CO₂ reduction of the

AS2M and DC2M processes integrated with different H₂ production technologies, such as steam methane reforming with/without carbon capture (SMR, SMR+CCS), coal gasification with/without carbon capture (CG, CG+CCS), and biomass gasification with/without carbon capture (BG, BG+CCS). The parameters for conventional technologies are adopted from previous studies [50, 51].

Compared to the renewable ELZs, the AS2M and DC2M processes using conventional H₂ production technologies produce cheaper methanol (A7: \$1.48/kg, D7: \$1.02/kg). Among alternatives, SMR is the cheapest among all conventional H₂ technologies, while BG+CCS is the most expensive in both the AS2M and DC2M processes. However, SMR reduces the CO₂ reduction impact, whereas BG+CCS has the largest CO₂ reduction effect than other H₂ production technologies. CG gives the worst CO₂ reduction effect compared to the other conventional technologies. Interestingly, the additions of CCS to the SMR, CG, and BG increase the total costs, although they largely improve the CO₂ reduction effect of conventional technologies. For example, the CO₂ reduction effect of the BG was improved by around five times by adding CCS. The production cost of methanol from the AS2M process with conventional H₂ technologies is still higher than the market price (c.f., MeOH market price: \$0.23–\$0.83/kg MeOH [52]), except for the cases using SMR, due to the complex process scheme and tight operation conditions of the AS2M process. In contrast, using conventional H₂ technologies improves the economic performance of the DC2M process. In particular, the

DC2M process, coupled with SMR, produces the cheapest methanol (i.e., \$0.41/kg), which is competitive with the market price of methanol.

6. Conclusions

In this work, we proposed new methanol synthesis processes that use waste CO₂ and renewable H₂ as feedstock. We analyzed the technoeconomic and environmental capabilities of the proposed processes by developing a process model using a commercial simulator. Specifically, two distinct processes, i.e., AS2M and DC2M, were proposed and compared to conventional CO₂-to-MeOH processes. The AS2M process produces methanol from syngas (a mixture of CO and H₂) after converting CO₂ to CO via reverse water-gas shift and separating CO from the mixture through pressure swing adsorption, while the DC2M process directly synthesizes methanol from CO₂ using the direct hydrogenation process. The optimal process configuration and operating conditions of the two processes were optimized for maximizing methanol production. The technical and economic capabilities of the two novel processes were examined using four evaluation criteria: carbon efficiency, energy efficiency, CO₂ reduction, and unit production cost. Our findings showed that both the AS2M and DC2M processes could achieve better technical and environmental performances compared to the existing methanol synthesis processes using CO₂. The H₂ production cost was found to be the most important factor in determining the economic feasibility of the processes. The AS2M process had the highest methanol yields for the given amount of CO₂ feed, while the DC2M process had the lowest unit production cost at \$1.02/kg because of its simple process design and easy operating conditions. In this study, we also identified that the proposed two processes can have economic competitiveness depending on the H₂ production technologies.

Nomenclature

Indices/Sets/Subsets

- $i \in I$: Compounds {CO, CO₂, H₂, H₂O, MeOH}
 $k \in K$: Utilities {Ec, Ht, Cw, Iw, Ad, Ab}
 Ec: electricity
 Ht: heat
 Cw: cooling water
 Iw: ionized water
 Ad: adsorbent for PSA
 Ab: absorbent for CO₂ recovery
 K_u : Electricity and heat utilities {Ec, Ht}, $K_u \subseteq K$.

Parameters

- α : Capital charge factor (yr⁻¹)
 s : Interest rate (%)
 n : Plant lifetime (yr)
 δ : Operability of the process (h/yr)
 τ_{MeOH}^C : Amount of carbon element in methanol (kmol of carbon/kmol of methanol)
 $\tau_{\text{CO}_2}^C$: Amount of carbon element in CO₂ (kmol of carbon/kmol of CO₂)

- π_{MeOH} : Low heating value of methanol (MJ/kg MeOH)
 M_{MeOH} : Molar weight of methanol (kg/kmol)
 M_{CO_2} : Molar weight of CO₂ (kg/kmol)
 e_{heat} : CO₂ emission factor of heat utility (kg of CO₂/MJ of heat)
 e_{elect} : CO₂ emission factor of electricity utility (kg of CO₂/MJ of electricity)
 θ_k : Unit price of utility k (\$/MJ or \$/kg)
 θ_{O_2} : Selling price of oxygen (\$/kg of O₂).

Continuous Variables

- C_{MeOH} : Total number of carbons in methanol (kmol of carbon/h)
 C_{CO_2} : Total number of carbons in CO₂ (kmol of carbon/h)
 F_{MeOH} : Mole flow of methanol (kmol of methanol/h)
 F_{CO_2} : Mole flow of CO₂ (kmol of CO₂/h)
 F_{O_2} : Mole flow of oxygen (kmol of O₂/h)
 E_{MeOH} : Chemical energy of methanol produced in the process (MJ/yr)
 E_k : Amount of utilities $k \in K_u$ consumed in the process (MJ/yr)
 A_k : Amount of utilities $k \in K \setminus K_u$ consumed in the process (kg/yr)
 DE: Direct CO₂ emissions (kg of CO₂/h)
 IE: Indirect CO₂ emissions (kg of CO₂/h)
 CE: Carbon efficiency (%)
 EE: Energy efficiency (%)
 CR: CO₂ reduction (kg CO₂/kg MeOH)
 TPC: Total production cost (\$/yr)
 UPC: Unit production cost (\$/kg MeOH)
 CC: Capital cost (\$/yr)
 OC: Operating cost (\$/yr)
 FC: Fixed operating cost (\$/yr)
 VC: Variable operating cost (\$/yr)
 BC: Byproduct credit (\$/yr).

Data Availability

The manuscript includes all the data (e.g., parameters, simulation data, and analysis results).

Conflicts of Interest

The authors declare that they have no conflict of interest.

Authors' Contributions

Seolhee Cho and Thai Ngan Do contributed equally to this work.

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Supplementary Materials

The supplementary material contains additional information about the process flow diagram and major stream information of the o-CAMERE and m-CAMERE processes (Figure S1 and S2, respectively). In addition, the techno-economic and environmental analysis results (Table S1) can be found in the supplementary materials. (*Supplementary Materials*)

References

- [1] V. Masson-Delmotte, P. Zhai, H. O. Pörtner et al., *IPCC Special Report: Global Warming of 1.5°C*, 2018.
- [2] P. Jaramillo, W. M. Griffin, and S. T. McCoy, "Life cycle inventory of CO₂ in an enhanced oil recovery system," *Environmental Science & Technology*, vol. 43, no. 21, pp. 8027–8032, 2009.
- [3] S. Cho, S. Kim, and J. Kim, "Life-cycle energy, cost, and CO₂ emission of CO₂-enhanced coalbed methane (ECBM) recovery framework," *Journal of Natural Gas Science and Engineering*, vol. 70, article 102953, 2019.
- [4] S. M. Jarvis and S. Samsatli, "Technologies and infrastructures underpinning future CO₂ value chains: a comprehensive review and comparative analysis," *Renewable and Sustainable Energy Reviews*, vol. 85, pp. 46–68, 2018.
- [5] F. D. Meylan, V. Moreau, and S. Erkman, "CO₂ utilization in the perspective of industrial ecology, an overview," *Journal of CO₂ Utilization*, vol. 12, pp. 101–108, 2015.
- [6] C. Hepburn, E. Adlen, J. Beddington et al., "The technological and economic prospects for CO₂ utilization and removal," *Nature*, vol. 575, no. 7781, pp. 87–97, 2019.
- [7] A. Saravanan, D. V. Vo, S. Jeevanantham et al., "A comprehensive review on different approaches for CO₂ utilization and conversion pathways," *Chemical Engineering Science*, vol. 236, article 116515, 2021.
- [8] A. D. N. Kamkeng, M. Wang, J. Hu, W. Du, and F. Qian, "Transformation technologies for CO₂ utilisation: current status, challenges and future prospects," *Chemical Engineering Journal*, vol. 409, article 128138, 2021.
- [9] A. Otto, T. Grube, S. Schiebahn, and D. Stolten, "Closing the loop: captured CO₂ as a feedstock in the chemical industry," *Energy & Environmental Science*, vol. 8, no. 11, pp. 3283–3297, 2015.
- [10] D. T. Whipple and P. J. A. Kenis, "Prospects of CO₂ utilization via direct heterogeneous electrochemical reduction," *Journal of Physical Chemistry Letters*, vol. 1, no. 24, pp. 3451–3458, 2010.
- [11] S. Saeidi, N. A. S. Amin, and M. R. Rahimpour, "Hydrogenation of CO₂ to value-added products—a review and potential future developments," *Journal of CO₂ Utilization*, vol. 5, pp. 66–81, 2014.
- [12] S. Najari, G. Gróf, S. Saeidi, and F. Gallucci, "Modeling and optimization of hydrogenation of CO₂: estimation of kinetic parameters via artificial bee colony (ABC) and differential evolution (DE) algorithms," *International Journal of Hydrogen Energy*, vol. 44, pp. 4630–4649, 2019.
- [13] S. Poto, F. Gallucci, and M. Fernanda Neira d'Angelo, "Direct conversion of CO₂ to dimethyl ether in a fixed bed membrane reactor: influence of membrane properties and process conditions," *Fuel*, vol. 302, article 121080, 2021.
- [14] Y. Khojasteh-Salkuyeh, O. Ashrafi, E. Mostafavi, and P. Navarri, "CO₂ utilization for methanol production; part I: process design and life cycle GHG assessment of different pathways," *Journal of CO₂ Utilization*, vol. 50, article 101608, 2021.
- [15] A. Rafiee, K. Rajab Khalilpour, D. Milani, and M. Panahi, "Trends in CO₂ conversion and utilization: a review from process systems perspective," *Journal of Environmental Chemical Engineering*, vol. 6, no. 5, pp. 5771–5794, 2018.
- [16] J. C. Abanades, E. S. Rubin, M. Mazzotti, and H. J. Herzog, "On the climate change mitigation potential of CO₂ conversion to fuels," *Energy & Environmental Science*, vol. 10, no. 12, pp. 2491–2499, 2017.
- [17] S. Michailos, S. McCord, V. Sick, G. Stokes, and P. Styring, "Dimethyl ether synthesis via captured CO₂ hydrogenation within the power to liquids concept: a techno-economic assessment," *Energy Conversion and Management*, vol. 184, pp. 262–276, 2019.
- [18] M. Aresta, A. Dibenedetto, and E. Quaranta, "State of the art and perspectives in catalytic processes for CO₂ conversion into chemicals and fuels: the distinctive contribution of chemical catalysis and biotechnology," *Journal of Catalysis*, vol. 343, pp. 2–45, 2016.
- [19] A. González-Garay, M. S. Frei, A. Al-Qahtani, C. Mondelli, G. Guillén-Gosálbez, and J. Pérez-Ramírez, "Plant-to-planet analysis of CO₂-based methanol processes," *Energy & Environmental Science*, vol. 12, no. 12, pp. 3425–3436, 2019.
- [20] J. Kim, C. A. Henao, T. A. Johnson et al., "Methanol production from CO₂ using solar-thermal energy: process development and techno-economic analysis," *Energy & Environmental Science*, vol. 4, no. 9, pp. 3122–3132, 2011.
- [21] J. Kim, T. A. Johnson, J. E. Miller, E. B. Stechel, and C. T. Maravelias, "Fuel production from CO₂ using solar-thermal energy: system level analysis," *Energy & Environmental Science*, vol. 5, no. 9, pp. 8417–8429, 2012.
- [22] J. Kim, J. E. Miller, C. T. Maravelias, and E. B. Stechel, "Comparative analysis of environmental impact of S2P (sunshine to petrol) system for transportation fuel production," *Applied Energy*, vol. 111, pp. 1089–1098, 2013.
- [23] T. N. Do, C. You, and J. Kim, "A CO₂ utilization framework for liquid fuels and chemical production: techno-economic and environmental analysis," *Energy & Environmental Science*, vol. 15, no. 1, pp. 169–184, 2022.
- [24] Project ALIGN-CCUS <https://www.rwe.com/en/research-and-development/rwe-innovation-centre/e-fuels/align-ccus/>.
- [25] P. Moser, G. Wiechers, S. Schmidt et al., *ALIGN-CCUS: production of dimethyl ether from CO₂ and its use as energy carrier - results from the CCU demonstration plant March 2021*, Elsevier BV, 2021.
- [26] S. Troy, P. Zapp, W. Kuckshinrichs et al., *Life cycle assessment for full chain CCU demonstration in the align-CCUS project - dimethyl ether and polyoxymethylen dimethyl ethers production from CO₂ and its usages in the mobility and electricity sectors March 2021*, Elsevier BV, 2021.
- [27] F. G. Albrecht, D. H. König, N. Bauks, and R. U. Dietrich, "A standardized methodology for the techno-economic evaluation of alternative fuels - a case study," *Fuel*, vol. 194, pp. 511–526, 2017.
- [28] N. Von Der Assen, J. Jung, and A. Bardow, "Life-cycle assessment of carbon dioxide capture and utilization: avoiding the pitfalls," *Environmental Sciences*, vol. 6, no. 9, p. 2721, 2013.
- [29] C. Zhang, K. W. Jun, R. Gao, G. Kwak, and H. G. Park, "Carbon dioxide utilization in a gas-to-methanol process combined

- with CO₂/steam-mixed reforming: techno-economic analysis," *Fuel*, vol. 190, pp. 303–311, 2017.
- [30] Australia's National Science Agency, "Methanol," <https://www.csiro.au/en/work-with-us/ip-commercialisation/hydrogen-technology-marketplace/methanol>.
- [31] T. Xin, C. Xu, Y. Liu, and Y. Yang, "Thermodynamic analysis of a novel zero carbon emission coal-based polygeneration system incorporating methanol synthesis and Allam power cycle," *Energy Conversion and Management*, vol. 244, article 114441, 2021.
- [32] D. Mignard and C. Pritchard, "Processes for the synthesis of liquid fuels from CO₂ and marine energy," *Chemical Engineering Research and Design*, vol. 84, no. 9, pp. 828–836, 2006.
- [33] D. Mignard, M. Sahibzada, J. M. Duthie, and H. W. Whittington, "Methanol synthesis from flue-gas CO₂ and renewable electricity: a feasibility study," *International Journal of Hydrogen Energy*, vol. 28, no. 4, pp. 455–464, 2003.
- [34] Y. A. Daza and J. N. Kuhn, "CO₂ conversion by reverse water gas shift catalysis: comparison of catalysts, mechanisms and their consequences for CO₂ conversion to liquid fuels," *RSC Advances*, vol. 6, no. 55, pp. 49675–49691, 2016.
- [35] O. S. Joo, K. D. Jung, I. Moon et al., "Carbon dioxide hydrogenation to form methanol via a reverse-water-gas-shift reaction (the CAMERE process)," *Industrial and Engineering Chemistry Research*, vol. 38, no. 5, pp. 1808–1812, 1999.
- [36] D. S. Marlin, E. Sarron, and Ó. Sigurbjörnsson, "Process advantages of direct CO₂ to methanol synthesis," *Frontiers in Chemistry*, vol. 6, p. 446, 2018.
- [37] X. An, Y. Zuo, Q. Zhang, and J. Wang, "Methanol synthesis from CO₂ hydrogenation with a Cu/Zn/Al/Zr fibrous catalyst," *Chinese Journal of Chemical Engineering*, vol. 17, no. 1, pp. 88–94, 2009.
- [38] A. A. Kiss, J. J. Praght, H. J. Vos, G. Bargeman, and M. T. de Groot, "Enhanced process for methanol production by CO₂ hydrogenation," *Computer Aided Chemical Engineering*, vol. 38, pp. 985–990, 2016.
- [39] A. A. Kiss, J. J. Praght, H. J. Vos, G. Bargeman, and M. T. de Groot, "Novel efficient process for methanol synthesis by CO₂ hydrogenation," *Chemical Engineering Journal*, vol. 284, pp. 260–269, 2016.
- [40] T. N. Do and J. Kim, "Process development and techno-economic evaluation of methanol production by direct CO₂ hydrogenation using solar-thermal energy," *Journal of CO₂ Utilization*, vol. 33, pp. 461–472, 2019.
- [41] S. Cho, C. Kim, and J. Kim, "Techno-economic assessment and early-stage screening of CO₂ direct hydrogenation catalysts for methanol production using knowledge-based surrogate modeling," *Energy Conversion and Management*, vol. 244, article 114477, 2021.
- [42] Aspen PlusVersion 11, Aspen Technology Inc, n.d. <http://www.aspentech.com/>.
- [43] I. D. G. Chaves, J. R. G. López, J. L. G. Zapata, A. L. Robayo, and G. R. Niño, *Process Analysis and Simulation in Chemical Engineering*, Springer, Cham, 2015.
- [44] Y. I. Lim, J. Choi, H. M. Moon, and G. H. Kim, "Techno-economic comparison of absorption and adsorption processes for carbon monoxide (CO) separation from Linde-Donawitz gas (LDG)," *Korean Chemical Engineering Research*, vol. 54, no. 3, pp. 320–331, 2016.
- [45] E. Taibi, R. Miranda, W. Vanhoudt, T. Winkel, J. C. Lanoix, and F. Barth, *Hydrogen from renewable power: technology outlook for the energy transition*, IRE, 2018.
- [46] U. Fritsche and K. Schmidt, *Global emission model of integrated systems (GEMIS) manual*, Institute for Applied Ecology, 2008.
- [47] International Renewable Energy Agency, *Renewable power generation costs in 2017*, IRENA, Abu Dhabi, 2011.
- [48] S. Kim, D. Ko, J. Mun, T. H. Kim, and J. Kim, "Techno-economic evaluation of gas separation processes for long-term operation of CO₂ injected enhanced coalbed methane (ECBM)," *Korean Journal of Chemical Engineering*, vol. 35, no. 4, pp. 941–955, 2018.
- [49] W. Won, H. Kwon, J. H. Han, and J. Kim, "Design and operation of renewable energy sources based hydrogen supply system: technology integration and optimization," *Renewable Energy*, vol. 103, pp. 226–238, 2017.
- [50] A. Al-Qahtani, B. Parkinson, K. Hellgardt, N. Shah, and G. Guillen-Gosalbez, "Uncovering the true cost of hydrogen production routes using life cycle monetisation," *Applied Energy*, vol. 281, article 115958, 2021.
- [51] A. Zapantis, *Blue hydrogen*, Global CCS Institute, Melbourne, Australia, 2021.
- [52] Methanex Corporation <https://www.methanex.com/our-business/pricing>.